

Nikhil Sai Ramini

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SUMMARY

Software Engineer with a strong foundation in Python, Java, C, C++, SQL and over 2+ years of hands-on experience transitioning from Data Science to backend-focused development. Proven track record of building and optimizing scalable data pipelines, cloud-integrated systems, and high-performance backend services. Successfully developed and deployed efficient IoT solutions with a 30% increase in data transmission efficiency. Experienced in CI/CD pipelines, RESTful APIs, and containerized deployments using AWS and Docker. Passionate about designing clean, maintainable code and delivering robust, AI-driven software products in fast-paced environments.

EDUCATION

Ohio University

Master of Science, Information & Telecommunication Systems

Aug 2023 - Dec 2024

Avanthi Scientific And Technological Research Academy

Bachelor of Technology, Computer Science and Engineering

Jul 2019 - May 2023

- **Achievements:** Organized and participated in peer-led coding sessions and algorithm practice groups, mentoring juniors on introductory programming and helping classmates strengthen their Python, Java, and C skills through mock interviews and coding challenges

WORK EXPERIENCE

Alpha Recon

Software Engineer

Feb 2025 - Present

- Built and deployed scalable machine learning pipelines (TF-IDF, SMOTE, Random Forest/XGBoost) to classify security incidents across U.S. states using geospatial, temporal, and textual features.
- Integrated Retrieval-Augmented Generation (RAG) to automate and personalize state-specific security intelligence reports using templated summaries and LLMs.
- Fine-tuned large language models (LLMs) and integrated them with LangChain to enable contextual retrieval, conversational reasoning, and automated report drafting.
- Developed AI agents capable of executing multi-step reasoning tasks, integrating external APIs, and autonomously generating domain-specific intelligence outputs.
- Designed and maintained efficient data pipelines for automated data processing, model training, and inference, leveraging SQL databases where applicable.
- Collaborated with cross-functional teams including engineers and product managers to align technical solutions with business needs.
- Authored clear, reproducible technical documentation and presented key findings to technical and non-technical stakeholders.

Ohio University

IOT- Development, Programming, and Deployment/ Graduate Assistant

Apr 2024 - Dec 2024

Athens, Ohio, US

- Developed, programmed, and deployed LoRaWAN IoT systems, enhancing communication efficiency and supporting integrated system operations.
- Configured and managed LoRaWAN systems by integrating Grove devices, Raspberry Pi with RAK HAT, sensors, gateways, and GPS modules for optimized IoT communication.
- Utilized Python for system configuration and data processing along with Arduino (C++) for microcontroller programming involving sensor integration, serial communication, and module configuration.
- Optimized data transmission efficiency by 30%, significantly improving overall system performance and reliability.
- Conducted comprehensive functional, performance, power, and stress testing to resolve hardware-software issues and enhance system robustness.
- Deployed a robust LoRaWAN system capable of handling real-time communication and data exchange between sensors and servers while focusing on user experience.

Ohio university

Technical Support/ Graduate Assistant

Aug 2023 - Mar 2024

Athens, Ohio, US

- Provided technical troubleshooting and maintenance for lab computers, printers, and network systems, ensuring optimal functionality and minimal downtime.
- Conducted troubleshooting and tier-1 support across LAN/WAN devices using tools like Wireshark, while maintaining clear documentation for all changes.
- Assisted students, faculty, and staff with resolving technical issues, delivering hands-on support and user training as needed.
- Resolved tickets in the OIT department, addressing a range of technical issues and ensuring timely solutions.
- Configured and deployed new workstations, peripherals, and network setups, ensuring seamless integration and operation.
- Documented technical issues, resolutions, and procedures to create a knowledge base and improve support efficiency.
- Streamlined lab operations by implementing system updates, monitoring resource usage, and ensuring compliance with institutional policies.

PROJECTS

BARN Zero Trust Model

- Designed and implemented a Zero Trust network architecture based on the BARN model to enhance organizational security. Utilized principles of continuous verification, least privilege access, and micro-segmentation to mitigate internal and external threats.

IPv6 Deployment for Ohio University

- Led the planning and deployment of IPv6 within the university network in Infoblox. Conducted address planning, DNS configuration, and router interface setups to ensure seamless coexistence with IPv4 and future-proof network infrastructure.

ML Techniques for Cyber Attacks Detection

- Developed a cyber attack detection system using machine learning algorithms such as decision trees and neural networks. Implemented the solution using Python, scikit-learn, and TensorFlow to classify network anomalies with high accuracy.

Stock Market Analysis and Prediction

- Built a predictive model for stock price movement using historical market data and technical indicators. Employed Python with libraries like pandas, scikit-learn, and matplotlib for data analysis, feature engineering, and model evaluation.

COP FLIX - Cinematic Virtual Reality Streaming Service

- Created a VR-based streaming platform offering immersive cinematic experiences. Engineered the system with Unity and C#, integrating 360-degree video, spatial audio, and user interactivity for an engaging virtual entertainment solution.

SKILLS

- **Programming & Development:** Python, C, Java, C++, R, JavaScript, HTML, CSS, Powershell, User Experience Focus, React, Objective-C, Data Science, Object-Oriented Programming (OOP), Node.js, Product Development, ETL, Shell Scripting, Bash, Object Oriented Programming
- **Data & Cloud Engineering:** SQL, MYSQL, Data Engineering, MongoDB, Data Pipelines, AWS, Cloud Infrastructure, CI/CD Workflows, Azure, Docker, Cloud Computing, GCP, Kubernetes, Terraform, Ansible, Git
- **Artificial Intelligence & Cyber Security:** AI, Machine Learning, RLHF, Generative AI, Prompt Engineer, LLMs, SIEM, Threat Detection & Analysis, Security Operations (SOC Fundamentals), Vulnerability Assessment, Risk Assessment & Management, Network Security, Cryptography Basics, Security Policies & Compliance Awareness
- **Systems & Tools:** VoIP, IoT Systems Development, Programming, Deployment, DHCP, DNS, IPv4, IPv6, SDN, Infoblox, 5G Wireless, Distributed Systems, Code Review, Software Integration, Technical Documentation, Troubleshooting, Microcontrollers, Tableau, Linux Kernel, Backend Development, Containerization
- **Networking:** Network Architecture Design, TCP/IP, OSI Model, Switching & Routing, Ethernet, Optical Networking (Fiber, Multimode, Single-mode), VPN, QoS (Quality of Service), Network Topology, Network Planning & Optimization, Troubleshooting / Problem Resolution, VLANs, Packet Analysis Tools (Wireshark), Network Devices & Appliances (Routers, Switches, Modems), SDN, REST APIs, VMware, Wireshark, Active Directory
- **Operating Systems:** Windows, macOS, Linux (Ubuntu, Raspbian), Linux Kernel
- **Tools & Technologies:** VMware, Active Directory, Remote Desktop, SCCM, Jira, Infoblox, LoRaWAN
- **Hardware:** Raspberry Pi, Arduino, Desktop/Laptop Support, Peripheral Devices, Microcontrollers
- **Soft Skills:** Problem-solving, Technical Communication, Documentation, Project Coordination

CERTIFICATES & BADGES

- **Google Cloud Essentials:** Completed training in GCP services, including compute, storage, IAM, networking, and cloud monitoring.
- **Python Essentials 1&2 - Cisco:** Learned Python programming, data structures, OOP concepts, and scripting for automation and application development.
- **Python Certification - Kaggle:** Mastered core Python skills for data science, including data types, control flow, functions, and basic libraries like Pandas and NumPy.
- **Junior Cybersecurity Analyst - Cisco:** Developed practical skills in identifying threats, performing risk assessments, and understanding SOC operations and security policies.
- **Tata Group's Cybersecurity Analyst Job Stimulation:** Simulated real-world scenarios to analyze vulnerabilities, review logs, and recommend security enhancements in enterprise environments.
- **Introduction to Cybersecurity - Cisco:** Learned cybersecurity principles, threat vectors, cryptography basics, and best practices for protecting networks and data.
- **5G Wireless Certification - Telecommunications Education Center (TEC) :** Gained insights into 5G architecture, network slicing, edge computing, and the impact of 5G on IoT and enterprise systems.

ACHIEVEMENTS

- Earned a Gold Badge in Python on HackerRank, showcasing advanced problem-solving skills and expertise in Python programming
- Achieved Silver Badges in both C and Java on HackerRank, demonstrating proficiency in multiple programming languages and strong coding abilities.
- Designed and deployed a LoRaWAN-based IoT sensor network on Ohio University's Green Roof, enabling real-time environmental monitoring and providing data-driven insights to support climate research and sustainability initiatives.
- Best Student President Avanthi's Scientific And Technological Research Academy 2022-23